

Original Research

Capacitance, charge dynamics, and electrolyte-surface interactions in functionalized carbide-derived carbon electrodes

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Abstract

This study analyzed the dynamics of ionic liquid electrolyte inside of defunctionalized, hydrogenated, and aminated pores of carbide-derived carbon supercapacitor electrodes. The approach tailors surface functionalities and tunes nanoporous structures to decouple the influence of pore wall composition on capacitance, ionic resistance, and long-term cyclability. Quasi-elastic neutron scattering probes the self-diffusion properties and electrode-ion interactions of electrolyte molecules confined in functionalized pores. Room-temperature ionic liquid interactions in confined pores are strongest when the hydrogen-containing groups are present on the surface. This property translates into higher capacitance and greater ion transport through pores during electrochemical cycling. Unlike hydrogenated pores, aminated pores do not favorably interact with ionic liquid ions and, subsequently, are outperformed by defunctionalized surfaces.

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1. Introduction

A growing number of grid-storage, automotive, and portable electronics devices are relying on electrochemical capacitors to efficiently store and output electrical energy [1,2]. These energy storage systems, which are commonly known as supercapacitors, employ electrosorption of ions from electrolytes into carbon electrodes to assemble a double layer of charge on highly porous surfaces [3]. This approach relies on the high specific surface areas of electrode structures (500–2500 m² g^{−1}) and rapid ion dynamics (< 1 s) to maximize, respectively, the energy and power densities of such systems [4,5]. In addition to maximizing the capacitance, operating voltage window, and long-term cycle lives of these systems, many research efforts are

currently investigating the fundamental charge storage mechanism and decoupling key parameters that govern electrode–electrolyte interactions and dynamics. This crucial knowledge is required for future material and device design of high-performance supercapacitors.

Most prior work has investigated strictly the influence of pore size in carbon electrodes (typically in the 0.5–2.0 nm diameter range) [6] and the influence of the ion to pore-size ratio on capacitance [7]. These computation and experimental efforts mostly implemented a coarse-grained approach and assumed the pores and ions as, respectively, defect-free surfaces and hard spheres [8]. However, emerging research has also highlighted substantial differences between bulk (outside of particles) and confined electrolyte (inside of pores) [9]. In addition to pore size, surface functional groups (oxygen, nitrogen, halogen, etc.) significantly influence ion dynamics and densities in pores *via* intermolecular interactions and steric repulsions [10]. These electrostatically-driven effects are unique for specific ions and solvents, since each electrode–electrolyte combination features different dipole moments and

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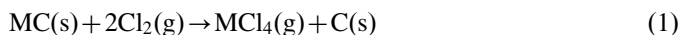
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molecular dimensions [11]. Although multiple existing studies have attempted to decouple these factors, they (1) mostly rely on MD simulations without experimental data support, (2) typically assume a planar graphene or external surface of a nanotube with no ion confinement, and (3) cannot fully correlate electrostatic interactions at the fluid–surface interface with specific capacitance values and rate handling abilities [7,12,13]. There is considerable practical interest in investigating these properties: H- and N-rich groups may improve electrochemical stability, enhance ion density in narrow pores, and add pseudocapacitive contributions.

Carbide-derived carbons (CDCs) exhibit high specific surface areas ($1000\text{--}2500\text{ m}^2\text{ g}^{-1}$) and exceptionally well-defined subnanometer ($d_{\text{av}} \approx 0.5\text{--}1.2\text{ nm}$) porosities [14]. Subsequently, as supercapacitor electrodes, they yield high gravimetric capacitance (above 150 F g^{-1} in organic electrolyte) and find use in both fundamental electrosorption studies and in commercial applications [15]. They are typically synthesized at $200\text{--}1200\text{ }^\circ\text{C}$ using the following reaction (M being an element such as Ti or Si) [16]:



These materials are typically annealed with H_2 (g) or NH_3 (g) at $600\text{ }^\circ\text{C}$ after synthesis to remove any residual Cl_2 [17]. Since the newly synthesized carbon structure has a chemically unstable surface (dangling bonds, lone pairs, and chemisorbed chlorides), these annealing steps selectively deposit functional groups and graft treatment-specific chemical species onto pore walls [18]. Typically, hydrogenation yields the C–H functional groups, while amination yields a heterogeneous mixture of C–N groups (C=NH, C–NH₂, quaternary nitrogen, etc.). Our previous approach stripped chemical groups from surfaces and increased graphitization in amorphous CDCs via $700\text{--}1800\text{ }^\circ\text{C}$, 10^{-6} Torr vacuum annealing [19]. These findings showed that defunctionalized, defect-free CDC surfaces exhibited lower capacitance but improved ion dynamics compared to unannealed CDCs with initial robust, heterogeneous surface chemistry compositions [20]. Although virtually every CDC electrode material tested in supercapacitor configurations has included one of these surface chemistry makeups (with hydrogen being most common), very few studies have addressed the influence of these functional groups on electrochemical performance. Most efforts have focused on extracting pseudocapacitive contributions to total charge densities by encouraging pH and surface charge-dependent interactions between pore wall chemical species and redox-active proton transfer electrolytes [21], such as aqueous KOH and H_2SO_4 . [22] However, novel room-temperature ionic liquid (RTIL) electrolytes do not engage in such H^+ transfer reactions [23,24]. Surface groups primarily influence RTIL interactions with pore walls via Coulombic coupling, steric interactions, and variations in pore filling densities [10]. Furthermore, conventional spectroscopy and elemental analysis techniques often struggle to properly quantify hydrogen (which is inaccessible to spectroscopy) and other functional groups on porous, semi-amorphous carbon such as activated carbon [18]. To improve the accuracy of computational analyses and properly tie them with experimental systems,

research efforts must combine electrochemical analysis of supercapacitor devices with comprehensive characterization of confined electrolytes and their fundamental dynamics inside of pores.

Our work compares the influence of three well-defined surface chemistries on pore walls of CDCs: defunctionalized (annealed), hydrogenated, and aminated. This approach focuses on electrodes with nearly-identical pore size distributions to properly decouple the influence of ion confinement from electrode surface composition. Electrochemical testing and intrinsic characterization evaluates interactions of divergently functionalized pores with a common ionic liquid electrolyte, 1-ethyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide ($[\text{EMIm}^+][\text{TFSI}^-]$). To compare the surface compositions and investigate resulting changes on ion dynamics, we rely on quasi-elastic neutron scattering (QENS) measurements using a high energy-resolution backscattering neutron spectrometer [25]. This approach measures diffusion of mobile species by analyzing broadening of the scattering signal in the range of energy transfers from $\pm 3.4\text{ }\mu\text{eV}$ (the resolution limit of the spectrometer) to $\pm 100\text{ }\mu\text{eV}$ [26]. QENS relies on the large incoherent neutron scattering cross-section of hydrogens compared to other elements to investigate self-diffusion and relative mobility properties of confined electrolyte ions. This measurement technique compares the molecules' respective relaxation times and fraction of elastic scattering in the total scattering signal [27]. Since all hydrogens in $[\text{EMIm}^+][\text{TFSI}^-]$ are attached to the imidazolium ring and alkyl chains, scattering from cations dominates the measured signal. Neutron-measured ion mobility correlates with their behavior under applied potentials in supercapacitor electrodes and shows the significant influences that H- and N-containing surface groups have as compared to defect-free pore walls. In addition to investigating capacitance and ion dynamics, we examine the influence of surface chemistry on ionic resistance and how that, in turn, affects long-term cyclability of these devices. Beyond CDC, these results are applicable to other carbon electrodes, such as activated carbon and graphene, which can be selectively functionalized using furnace and wet chemistry treatments.

2. Methods

2.1. Material synthesis

We synthesized carbide-derived carbons according to a previously developed procedure. [28] Titanium carbide (TiC) particles ($1\text{--}5\text{ }\mu\text{m}$, 99% pure, Alfa Aesar) were loaded into a quartz boat and placed in a quartz tube furnace. After flowing Ar gas purged the environment 1 h, the furnace ramped up to $800\text{ }^\circ\text{C}$, and the material was subjected to pure flowing Cl_2 gas for 6 h in the isothermal condition. The furnace then cooled down to $600\text{ }^\circ\text{C}$ in Ar gas, H_2 gas annealed the material for 3 h, and, finally, the furnace cooled down to room temperature. This material is hereafter referred to as “Initial CDC.”

2.2. Surface chemistry modification

The initial CDC material was defunctionalized at 1200 °C in high vacuum (10^{-6} Torr) for 8 hours using a graphite vacuum furnace (Solar Atmospheres) [20]. The material is hereafter referred to as “Annealed CDC.” Separately, we maximized the surface hydrogen content of a set of CDC materials by increasing the 600 °C H₂ annealing time to 6 h (this material is hereafter referred to as “Hydrogenated CDC”). Finally, flowing NH₃ (g) (instead of H₂ (g)) was used to anneal the materials after chlorine etching for 6 h at 600 °C and yielded nitrogen-rich CDC surfaces (referred to as “Aminated CDC”).

2.3. Materials characterization

We analyzed the pore structure of the materials and calculated their specific surface areas using isothermal gas adsorption. We used a Quadrasorb (Quantachrome Instruments) and conducted isothermal N₂ (g) sorption at 77 K. The Brunauer–Emmett–Teller (BET) specific surface area (SSA) was calculated from the 0.05 to 0.30 P/P₀ adsorption points [29], and the Quenched Solid Density Functional Theory (QSDFT, Quadrawin software package from Quantachrome) calculated pore size distributions. [30] A Q5000 SA Analyzer (TA Instruments) provided dynamic water vapor sorption measurements. The technique used isothermal H₂O uptake measurements at 25 °C by placing samples into Pt-coated balance pans and increasing relative humidity of the environment from 0% to 90%.

Raman spectroscopy analysis compared graphitization and vibrational modes of carbon with a 514 nm excitation Ar laser (5% power). A Renishaw inVia Spectrometer and an 1800 grooves cm⁻¹ grating collected and deconvoluted the spectra. A Q50 Thermal Analysis instrument was used to collect thermogravimetric analysis (TGA) in oxidizing (air-filled) and thermally decomposing (Ar-filled) environments. The samples were heated in quartz crucible on a balance pan from 100 °C to 850 °C at a 2 °C min⁻¹ temperature gradient.

To analyze the surface chemistry of the materials, we performed X-Ray Photoelectron Spectroscopy (XPS) analysis on a VersaProbe 5000 instrument employing a 100 μm monochromatic Al-Kα X-ray beam to irradiate the sample surface. The photoelectrons were collected by a 180° hemispherical electron energy analyzer. The samples were analyzed at a 45° takeoff angle between the sample surface and the path to the analyzer. Survey spectra were taken at the pass energy of 117.4 eV and with a step size of 0.500 eV. High-resolution spectra in the C 1 s, O 1 s, and N 1 s regions were taken at the pass energy of 23.5 eV and with a step size of 0.050 eV. The spectra were taken before and after sputtering with an Ar beam operating at 2 kV for 5 minutes. All binding energies were referenced to that of graphitic carbon at 284.4 eV. The quantification and peak fitting of the core-level spectra was performed using CasaXPS Version 2.3.16 RP 1.6. Prior to both the quantification and peak fitting the background contributions were subtracted using a Shirley function.

2.4. Quasi-elastic neutron scattering (QENS)

We conducted QENS measurements using the Backscattering Spectrometer (BASIS) BL-2 beamline of the Spallation Neutron Source at Oak Ridge National Laboratory. [25] Vacuum infiltration filled CDC pores with the ionic liquid 1-ethyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide ([EMIm⁺][TFSI⁻], Boulder Ionics, Inc.) electrolyte [9,31]. Details concerning our sample preparation steps are included in the Supporting Section, SI. The measurements were conducted in the $0.2 \text{ \AA}^{-1} < Q < 1.2 \text{ \AA}^{-1}$ range of scattering momentum transfer. The spectrometer operated at the 60 Hz frequency of the SNS. Quasi-elastic spectra was analyzed from the $\pm 100 \text{ } \mu\text{eV}$ energy transfer region [32]. The empty CDCs and filled CDCs for each surface chemistry configuration were analyzed with the instrument. Furthermore, since incoherent scattering from hydrogen dominated quasi-elastic scattering intensity, data analysis also calculated the percentage of elastically scattering (*i.e.* “immobilized” on the energy scale and time scale of our measurement) species in each sample. Measurements were conducted at ambient electrochemical conditions with no applied voltage. For each sample, we collected and analyzed data at 300 K, and subsequently cooled the sample down to the baseline temperature of 10 K to measure the sample-specific resolution spectrum used in the data fitting procedure. For IL-filled samples, the instrument also recorded the temperature dependence of the energy-resolved elastic scattering intensity from 10 K to 300 K at 5 K increments. This approach ruled out the presence of any bulk-like (10 nm in size or larger) [EMIm⁺][TFSI⁻] droplets that could potentially exhibit freezing–thawing behavior on the outside of CDC particles and validated both the RTIL infiltration approach and scattering measurements from confined ions.

2.5. Electrochemical testing

To correlate material properties and intrinsic ion dynamics with electrosorption rates and capacitance, we conducted standard electrochemical capacitor testing on each of the functionalized CDCs. Electrochemical cells used film electrodes by binding CDC powders with polytetrafluoroethylene (PTFE, Sigma-Aldrich) binder (95 wt% carbon, 5 wt% binder). The cells implemented a two-electrode pouch configuration with carbon-coated aluminum (Exopack) and Celgard as, respectively, the current collector and the separator. The sandwich cells, along with the neat [EMIm⁺][TFSI⁻] electrolyte, were assembled and cycled inside an Ar-filled glovebox. The electrochemical experiments were conducted with a VMP3 Potentiostat (Bio-Logic, Inc.) at room temperature. A 4-point probe measurement was compared with intrinsic electric conductivity of film electrodes (details in [Supporting Information](#)). Electrochemical impedance spectroscopy (EIS) was compared with ionic resistance and charge dynamics. The technique used a 10 mV oscillating potential amplitude centered at 0.0 V with a dampening 200 kHz to 10 mHz range frequency. Equivalent series resistance (ESR) derived from the

$Re(Z)$ value at which $Im(Z)$ impedance component became a zero value. The $Re(Z)$ difference between maximum charge transfer resistance (semi-circular region) and the regime above the $-45^\circ |Z|$ “knee frequency” compared ionic impedance contribution in each electrode. [33] To measure the capacitance, the cyclic voltammetry (CV) measurements were carried out between 0.0 and 2.5 V vertex potentials charged and discharged electrodes in the 2–250 $mV s^{-1}$ sweep rate range. The capacitance was derived from the discharge segments of each sweep, and the measurements were normalized by the mass of a single electrode and the intrinsic SSA of the CDC material. The equation below quantified the value of C_{sp} (V_0 and V_f denote, respectively 0.0 and 2.5 V, m_{WE} is the mass of a single working electrode, A_{sp} is the DFT-derived SSA, and V_w is the operating voltage window, and $\frac{dV}{dt}$ is the sweep rate):

$$C_{sp} = \frac{2}{A_{sp}} \int_{V_0}^{V_f} \frac{I}{\frac{dV}{dt} m_{WE} V_w} dV \quad (2)$$

Since the measurement collects the currents from both working and counter (symmetrical) electrodes, the measured C_{sp} value must include a factor of 2 in the numerator to derive capacitance for a single electrode. Finally, the galvanostatic

cycling at 5.0 $A g^{-1}$ for 10,000 cycles between 0 and 2.5 V on each cell evaluated the operating lifetimes of functionalized CDCs. This technique measured the coulombic efficiency (Q_{eff}) for each cycle and conducted cyclic voltammetry at 10 $mV s^{-1}$ before and after the long-duration cycling to calculate cycling-induced capacitance degradation.

3. Results and discussion

3.1. Material properties

3.1.1. Structural properties of CDCs

Our approach successfully yielded three porous carbon systems with divergent surface chemistries but nearly-identical porosities. As shown in Fig. 1a, each material featured a predominantly microporous structure, with most pores in the 0.61–0.66 nm range. A slight amount of mesoporosity ($d_{pore} \approx 1.2$ nm) was present for each material. Although the annealed CDC material was subjected to a 400 °C higher temperature than its hydrogenated or aminated counterparts, its annealing condition fell below the transition temperature at which significant graphitization of amorphous CDC would have set in (typically 1300–1400 °C) [19,34]. As

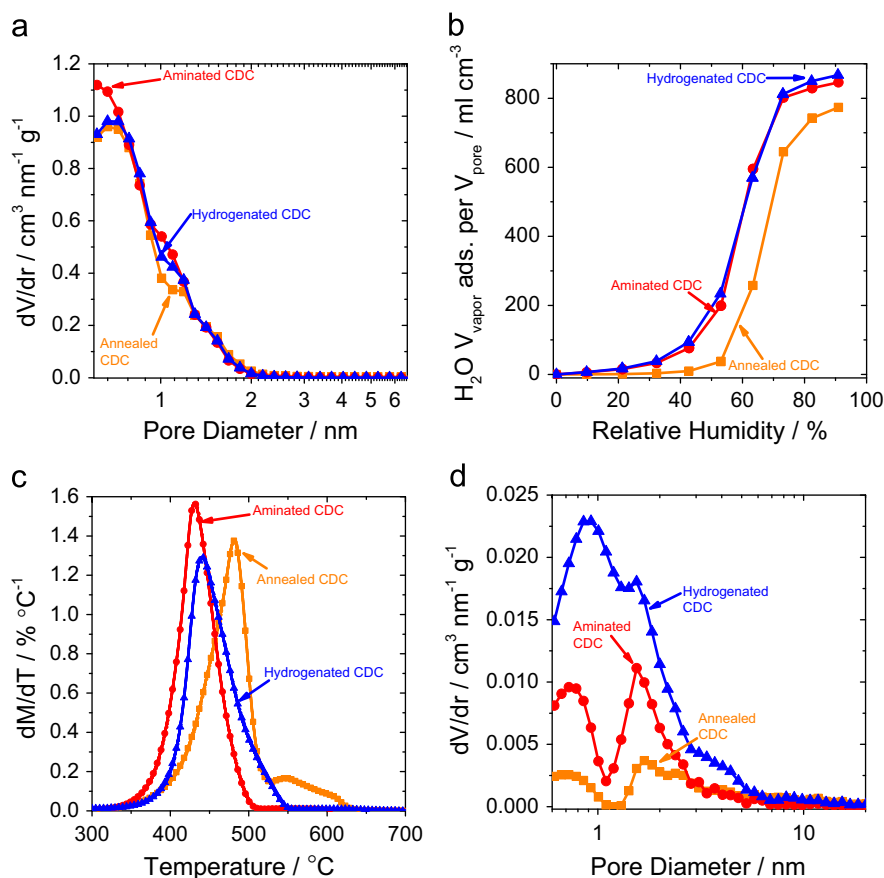


Fig. 1. (a) Pore size distribution for TiC-CDC-800 after 1200 °C vacuum annealing, H₂ annealing, and NH₃ annealing. (b) Dynamic water vapor sorption isotherms collected for the differently functionalized CDCs at 25 °C, demonstrating the amount of water vapor adsorbed at increasing relative humidity levels and normalized by empty pore volume of each material. (c) Derivative thermogravimetric analysis plot (dTG) for annealed and chemically modified CDCs from oxidation in air. (d) Porosimetry data on CDC materials after 80% of the pore volume of each was vacuum filled with [EMIm⁺][TFSI⁻].

shown in Fig. S1 in Supporting Information, although the annealed material is slightly more graphitized, all the three materials demonstrated fairly similar degree of carbon ordering. The SSA for all three materials (summarized in Table S1 in Supporting Information), which ranged from $1515 \text{ m}^2 \text{ g}^{-1}$ for hydrogenated CDC to $1728 \text{ m}^2 \text{ g}^{-1}$ for aminated CDC, was typical of TiC-CDCs synthesized at 800°C [28,35].

3.1.2. Surface chemistry of CDCs

The surface functional group composition of Annealed CDC pores became distinctly different from the Hydrogenated and Aminated ones. As shown in Fig. 1b, the chemistry of each significantly altered their respective hydrophobicity despite near-identical pore sizes. Hydrogen and nitrogen-containing groups equally favored significant H_2O vapor uptake, but the annealed pores were able to adsorb lower quantities of water and required higher vapor pressures for capillary action to overcome the unfavorable water–carbon interface. This was expected for defunctionalized carbon surfaces, which are primarily composed of hydrophobic graphitic carbon. Furthermore, the chemical groups on pore surfaces altered the thermal reactivity of the carbons in air. As shown in the TGA analysis in Fig. 1c, while hydrogenated and aminated CDCs combusted at lower temperatures (441°C and 431°C , respectively), defunctionalized CDCs burned away in air at a significantly higher temperature (481°C).

We relied on XPS analysis to determine the specific types of functional groups that were present on each CDC. The compositional analysis revealed a chemical makeup that we had anticipated, and agreed with prior characterization of differently annealed CDCs. The analysis of deconvoluted XPS spectra (Fig. 2) shows similar C 1s peaks for all three materials. These present peaks can be assigned to C–C (284.4 eV), C–H (285.2 eV), C–OH (286.3 eV), C=O (287.4 eV), and C–OOH (288.4 eV) species. There are likely C–NH and C–NH₂ species present in the Hydrogenated and Aminated CDCs as well (see below), however photoemission from those species overlaps with species present near $285\text{--}286 \text{ eV}$, and thus cannot be sufficiently de-convoluted. It should be noted that, as expected, the combined content of C–C and C–H species is $\sim 3.5\%$ higher for the Annealed CDC than for the Aminated or Hydrogenated CDCs. This is indicative of the removal of surface termination from this CDC. The analysis of the spectra taken in the N1s region (Fig. S2 in Supporting Information) reveals that the Annealed sample contains no nitrogen, the Hydrogenated sample contains two nitrogen species, and the Aminated sample contains three nitrogen species. Both the Hydrogenated and Aminated CDCs contain N 1s peaks near 400.2 eV and 398.7 eV , which correspond to NH and NH₂ groups, respectively. We note that the presence of these species in the Hydrogenated CDC likely arises from adsorbing N_2 from the air [36,37]. The third N

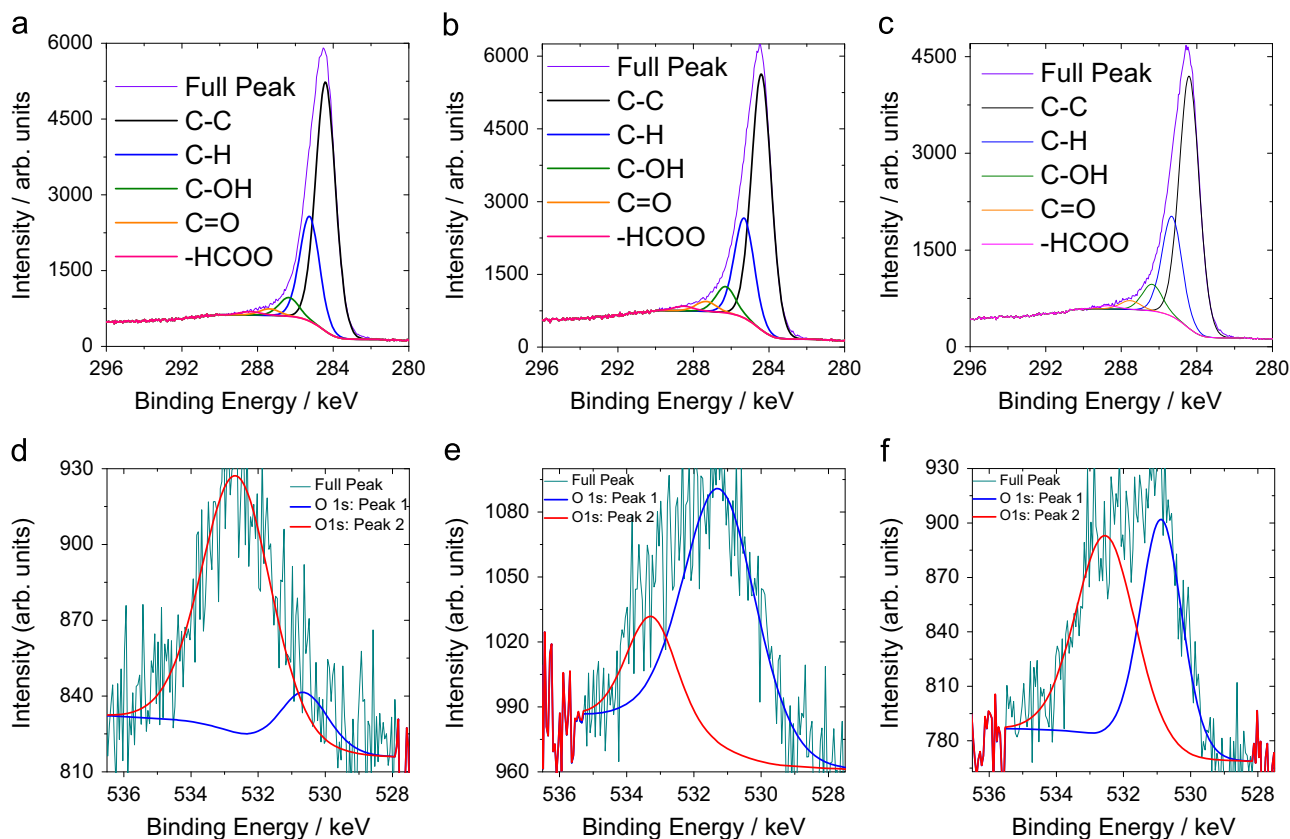


Fig. 2. High-resolution XPS spectra in the C1s region for (a) annealed CDC, (b) aminated CDC, and (c) hydrogenated CDC. Corresponding high-resolution XPS spectra in the O1s region are shown in (d), (e), and (f).

species present in the Aminated CDC occurs at 396.8 eV and is more consistent with that of a nitride. This peak may arise from titanium nitride which is formed from by reaction of the ammonia with remaining Ti that was not removed after the CDC synthesis. The fact that Ti was not identified as a contaminant on the survey spectra of this sample is indicative of the small amount present. The spectra in the O 1s region (Fig. 2) for the Annealed and Hydrogenated CDCs reveal the peaks for the species already identified in the C 1s spectra, including the peaks assigned to C–OH (~530.7 eV) and C=O/C–OOH (~532.6 eV) functional groups. Although it was never a part of the synthesis process, these oxygen peaks also stemmed from chemisorption of O₂ and H₂O from air [38]. The hydrogenated CDC contains more C–OH surface groups than C=O/C–OOH groups, which is expected given the hydrogen treatment. The O 1s spectrum of the Aminated sample also reveals two peaks, though they are shifted to 531.2 and 533.3 eV. The shift in these peaks is likely due to the inclusion of N–C–OH, and N–C=O/N–C–OOH functionalities, respectively, within these samples.

To summarize, as derived from the XPS spectra (Table 1), annealed CDC contained no nitrogen, while Aminated CDC contained (as expected) the greatest amount of nitrogen, and all three carbons contained small amounts of oxygen. In agreement with our results, the previous results have detected noticeable oxygen (6.1–10.1 wt%) quantities in aminated and hydrogenated porous carbons [39]. The chemical treatment (of any kind) of porous, semi-amorphous carbon leaves behind some unterminated/dangling bonds that likely become saturated with reactive species from the environment.

The conventional approach described above could not precisely quantify the amount of hydrogen on carbon surfaces; such analysis requires neutron activation spectroscopy or other similar techniques. TGA analysis heated CDC materials under an inert atmosphere to 850 °C (desorbing surface groups and leaving only carbon behind) and compared initial and final masses to quantify total surface chemistry compositions. This measurement (Table 1) derived significant amounts of hydrogen in hydrogenated CDCs; it had 10 wt% functional group content, as compared to 6.2 wt% for Aminated CDC and 4.0 wt% for Annealed CDC. Although other approaches, such as SOCl₂/NH₂(CH₂)₂NH₂ wet chemistry treatment, may selectively deposit –NH₂ groups on the carbon surfaces [40], such strong reactions are likely to significantly restructure/activate

the carbon materials and convolute porosities with surface chemistry effects.

3.2. Quasi-elastic neutron scattering: dynamics of confined ionic liquid

Based on the percentage of elastic scattering, as well as the lack of any observed transitions during heating of the sample from 10 K to 300 K, [EMIm⁺][TFSI[−]] was fully confined inside of micropores and left no bulk droplets on the surface. The analysis by scanning electron microscopy (SEM, Fig. S2 in Supporting Information) also shows no microscale droplets on the surfaces of particles. Subsequently, the confined hydrogen-containing [EMIm⁺] cations (in Filled CDC samples) and hydrogen-containing surface functional groups (in Empty CDC samples) overwhelmingly contributed to neutron scattering signals.

For all samples, the width of the scattering signal was found to be Q-independent, indicating localized, rather than long-range, mobility of ions on the time scale of our quasi-elastic measurements (tens to hundreds of picoseconds) and allowing summation of the data over the entire Q range of the experiment for data analysis. The resulting scattering signal as a function of neutron energy transfer, $I(E)$, was subsequently Fourier-transformed from the energy to the time space. The resulting incoherent scattering function, $I(t)$, contains information about single-particle microscopic dynamics of [EMIm⁺] cations. The decay of $I(t)$ as a function of time represents the loss of self-correlation in space with time, t , for a moving particle that was at the origin point at $t = 0$ [27]. There would be no time dependence for the $I(t)$ for a completely immobile particle, and neutron scattering from such particle would be fully elastic. Unlike for the vast majority of systems investigated by quasi-elastic neutron scattering, the $I(t)$ in our experiment could not be described satisfactorily by an exponential or a stretched exponential decay [41]:

$$I(t) = Ae^{-(\frac{t}{\tau})^\beta} \quad (3)$$

Instead, the best regression function to fit the $I(t)$ was a logarithmic decay model, expressed by the equation below (τ is the characteristic relaxation time):

$$I(t) = A - B * \ln\left(\frac{t}{\tau}\right) \quad (4)$$

This relationship is shown in Fig. 3a. The logarithmic decay model for $I(t)$, which is highly unusual in quasi-elastic scattering, underscores the low mobility of ions in narrow pores. Such functional form for the $I(t)$ was previously observed, surprisingly, only for such a disparate case as hydrated biomolecules [42]. Although the low mobility of cations, epitomized by the very slow logarithmic decay of the $I(t)$, precludes quantitative determination of the diffusion constants on the time scale of our quasi-elastic measurement, comparison of the incoherent scattering function among the samples is still informative.

Table 1

Elemental composition of differently functionalized CDCs, as determined by X-ray photoelectron spectroscopy. Trace impurities (Ti, Cl, and Fe) account for 0.0–0.1% of measured elemental compositions. The functional group content (column 4) was measured using thermogravimetric analysis of carbons in an inert (Ar-saturated) atmosphere.

Material	C (wt%)	O (wt%)	N (wt%)	Functional group content (%)
Annealed CDC	98.8	1.2	0	4.0
Aminated CDC	95.6	1.5	2.9	6.2
Hydrogenated CDC	95.2	2.1	2.6	10.9

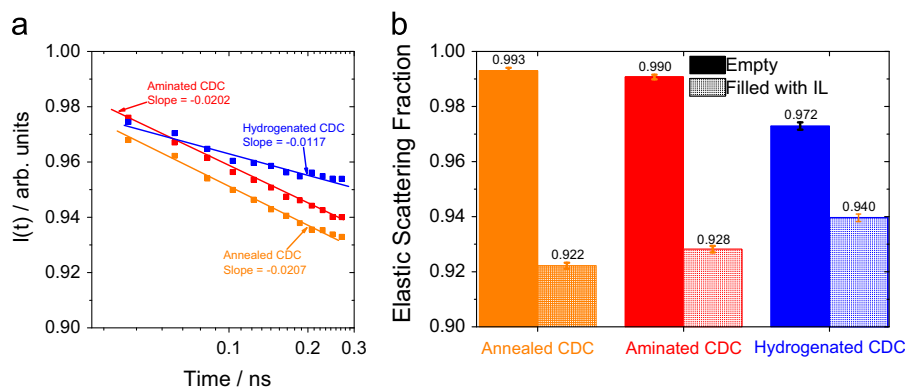


Fig. 3. (a) Logarithmic decay relationship between relaxation time and the incoherent scattering function (representing self-correlation in space) of mobile cations. (b) Fraction of elastic scattering for differently functionalized CDCs before and after IL loading. The fraction corresponds to the number of “immobilized” species in the sample that do not move appreciably on the time scale of our quasi-elastic measurement (tens to hundreds of picoseconds).

A linear regression analysis of the I vs. $\log(t)$ relationship derived the absolute value of the slope for hydrogenated CDC (0.0117) that was noticeably lower than for the Aminated CDC (0.0207) or annealed CDC (0.0202). Among the self-correlation functions for $[\text{EMIm}^+]$ cations in various environments, the $I(t)$ for the Hydrogenated CDC exhibited the slowest decay, suggesting that ions inside of H-terminated pores are least mobile.

Given the narrow pore diameter (~ 0.61 nm), large molecular diameters of both $[\text{EMIm}^+][\text{TFSI}^-]$, and our approach to only fill 80% of the pore volumes, we found that, in all likelihood, the ions inside of pores are in direct contact with the walls and the “middle” space in pores is left unoccupied. Furthermore, we expect the surface groups on CDCs that form strongest intermolecular interactions with the RTIL to draw the ions closer to pore walls and leave more void space around them. This was further confirmed using gas sorption measurements of filled CDCs (Fig. 1d): a percentage of micropores were left unoccupied. Hydrogenated Filled CDC demonstrated the most open space around confined ions. Given that pore filling was identical for each system, this surface exhibits the strongest intermolecular interactions between the ions and the hydrogenated surfaces that bind the ions. Therefore, the slowest dynamics for that system measured by QENS underscored the fact that the hydrogenated CDC pores interface is the most favorable for sorption of $[\text{EMIm}^+][\text{TFSI}^-]$.

Another way to extract information on the relative interaction strength of cations with CDC pore walls is to compare the apparent fraction of the elastic signal in the total neutron scattering spectrum. Stronger interaction with the pore walls leads to lower measurable mobility, and thus, to higher fraction of the elastic scattering. The discussion above concerned the dynamics of the mobile cations that manifests itself in the $I(t)$ decay; however, most cations in our samples were immobilized on the time scale of our quasi-elastic measurement (tens to hundreds of picoseconds). Coefficients of elastic scattering (Fig. 3b) for empty CDCs are primarily determined by hydrogen surface species, which are most “mobile” during neutron scattering events. The Annealed and Aminated samples show nearly-perfect elastic scattering (0.99), whereas the

lowest elastic scattering fraction for Hydrogenated CDC (0.972) further confirms the significantly greater amount of hydrogen on its pore walls. For the elastic scattering fractions measured for filled CDCs (where the signal is now dominated by the H-containing cations, whereas contributions from pore surface groups become relatively small), the situation is reversed: Hydrogenated CDC showed the most elastic scattering and, therefore, the most “immobilized” ions attached to pore’s surfaces. We can, therefore, conclude that the hydrogenated pore surface is most favorable to $[\text{EMIm}^+][\text{TFSI}^-]$ adsorption and attracts those ions the strongest. This finding fully agrees with our observation discussed above for those relatively few cations that exhibit measurable mobility on QENS time scales.

Our findings underscore the significance of intermolecular interactions between electrolyte ions and surface defects and functional groups on carbon electrodes. In addition to governing dynamics in confined pores, bonding forces also influence the thermal stability and degradation of these molecules. As shown in Fig. S4 in Supporting Information, $[\text{EMIm}^+][\text{TFSI}^-]$ confined in annealed pores thermally decomposes (into gas products such as SO_2 and CO_2) at a significantly lower temperature (393°C) than in its bulk state (437°C) [43]. Furthermore, the hydrogen and amine groups on surfaces, and their subsequent bonding interactions with $[\text{EMIm}^+][\text{TFSI}^-]$, decreased thermal decomposition temperature of the electrolyte even lower (371°C).

3.3. Electrochemical testing: capacitance, rate handling ability, and ion dynamics

We conducted cyclic voltammetry measurements and measured SSA-normalized capacitance for the differently functionalized CDCs with neat $[\text{EMIm}^+][\text{TFSI}^-]$ at room temperature. The results are shown in Fig. 4. It has been observed that even at low sweep rates (2 mV s^{-1} , Fig. 4a), the Hydrogenated electrode surface offered a higher capacitance than the Annealed and Aminated CDCs. At that sweep rate, the respective capacitances for the three materials are: 6.9, 6.5, and $5.9\text{ }\mu\text{F cm}^{-2}$. Although some ionic resistance slightly distorted

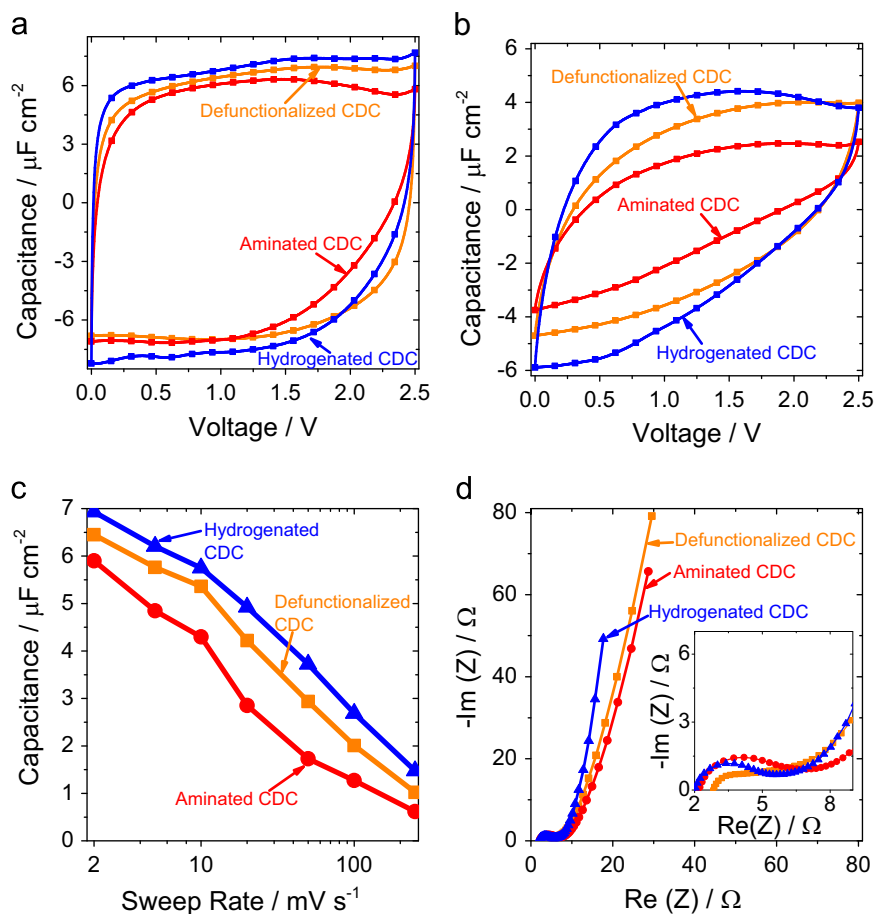


Fig. 4. Cyclic voltammograms of differently functionalized CDCs conducted at (a) 2 mV s^{-1} and (b) 50 mV s^{-1} with neat $[\text{EMIm}^+][\text{TFSI}^-]$ ionic liquid electrolyte. (c) Rate handling ability comparison for performance of CDCs in the $2.0\text{--}250 \text{ mV s}^{-1}$ sweep rate range. (d) Nyquist plot comparing electrochemical impedance for annealed, aminated, and hydrogenated CDC. The inset shows a zoomed-in high-frequency region that compares the ionic and charge transfer resistance contributions for the different electrode pore surfaces.

the rectangular (ideal) CV profile of each material, none of the materials exhibit distinguishable redox peaks at any sweep rates. Subsequently, neither the hydrogenated, nor aminated, nor side oxygen groups became involved in additional pseudocapacitive processes with $[\text{EMIm}^+][\text{TFSI}^-]$, and the charge storage that we measured was pure ion electrosorption.

The influence of surface groups on capacitance and ion dynamics became more apparent at 50 mV s^{-1} sweep rate (Fig. 4b): hydrogenated CDCs stored $3.7 \mu\text{F cm}^{-2}$, while Annealed CDCs and Aminated CDCs exhibited 2.9 and $1.7 \mu\text{F cm}^{-2}$, respectively. Higher-rate CVs showed greater ionic sieving that slightly affected Hydrogenated CDCs, but was increasingly significant for the other pore surface chemistries. Ionic resistance was especially apparent during the discharge segment of the CV sweep, suggesting that $[\text{TFSI}^-]$ anion is less easily electrosorbed and that it is less mobile in narrow pores. Ion sieving followed a similar trend for all three surface chemistries. However, it was significantly more pronounced for aminated CDC surfaces and hurt its capacitance the most. Subsequently, its nitrogen-rich pore wall chemistry is least favorable for $[\text{EMIm}^+][\text{TFSI}^-]$ electrosorption. This result contrasts with prior findings for aqueous electrolytes such as KOH, which demonstrated higher

capacitance (likely due to redox processes) in nitrogen-enriched porous carbons [22]. The small pore size of this model system promoted ion sieving and impeded ion mobility: the cross-sectional lengths and widths of $[\text{EMIm}^+]$ and $[\text{TFSI}^-]$ are, respectively, $0.43 \text{ nm} \times 0.69 \text{ nm}$ and $0.29 \text{ nm} \times 0.79 \text{ nm}$. Subsequently, CDC pores allow ions some translational motion but insufficient degree of freedom for unrestricted rotational motion. We implemented this condition to ensure that all ions will be in direct contact with pore walls, and that the results will not be convoluted with any void space for the electrolyte to adopt bulk-like behavior.

Hydrogenated CDC offered greater capacitance across the entire sweep range (Fig. 4c). Its rate handling ability and capacitance decay at higher cycling rates parallels the behavior of defunctionalized CDCs. However, the capacitance of aminated CDC decayed at a somewhat faster rate above 10 mV s^{-1} . We examined the mobility of ions across the $2\text{--}250 \text{ mV s}^{-1}$ sweep range and used the Randles–Sevcik equation to evaluate the diffusion limitation in each system [44]:

$$\log(I) = RS \log(v) \quad (5)$$

This equation finds the Randles–Sevcik (RS) coefficient using the peak discharge currents (I) for given CV sweep rates

(v). The results, along with key electrochemical parameters of different electrode materials, are summarized in Table 2. The RS coefficient approaches 1 for systems that behave like an ideal capacitor and 0.5 for systems with significant diffusion limitations. The values were almost identical for Hydrogenated and Annealed CDC (0.80 and 0.79, respectively), the Aminated CDC value (0.69) underscored substantial ion mobility restrictions. This relationship directly corresponds to the rate handling analysis.

Electrosorption-induced mobility of confined ions featured a unique correlation with neutron scattering-induced mobility of $[\text{EMIm}^+][\text{TFSI}^-]$. In the former case, we notice that hydrogenation of CDCs improved ion mobility and minimized ionic resistance better than the other two surface treatments. This relationship is supported by the EIS-derived Nyquist plot in Fig. 4d. We notice that the Equivalent Series Resistance (ESR), as well as the charge transfer resistance (semi-circular region), are identical for all three materials. However, hydrogenated CDC exhibits lower ionic impedance, whereas Annealed CDC and Aminated CDC feature depressed charge mobilities at low voltage oscillation frequencies (the f vs. C'' relationship is shown in Fig. S5 in Supporting Information) [45]. QENS also highlighted a significant deviation of RTIL mobility in hydrogenated CDC, but neutron scattering showed primarily “immobilized” ions in that system. Therefore, these slow neutron-measured dynamics of $[\text{EMIm}^+][\text{TFSI}^-]$ primarily highlight a favorable interface between the ionic liquid and the hydrogen-rich electrode surface. Under steady-state potential (including $V=0$) conditions, the intermolecular interactions between surface groups and the RTIL draw ions closer to the surface. However, this allows the ions to slide past the surface and past each other with less resistance during

electrochemical cycling and demonstrate a higher capacitance compared to Annealed CDCs. Aminated CDCs exhibited an opposite effect: those functional groups appear to repel $[\text{EMIm}^+][\text{TFSI}^-]$ ions and impede their transport and exchange in narrow pores. Our findings, which demonstrate the uniqueness of each carbon–electrolyte interface, agree with recent results that highlight distinct electrosorption behaviors for ionic liquids in nitrogen-rich carbon pores [46].

3.4. Long-term cycling and electrochemical stability

The influence of surface functional groups on cyclability of each material was evaluated. And the material was cycled at a high current loading rate (5.0 A g^{-1}) for 10,000 cycles to discern any evidence of supercapacitor degradation. The changes in coulombic efficiency (shown in Fig. 5a) demonstrated that the Hydrogenated CDC operated with the greatest degree of reversibility and stability throughout the entire cycling regime. It has been found that its Q_{eff} maintained 75% of its initial value, whereas the values for Annealed CDC and Aminated CDC declined to 65% and 57% of their respective starting efficiencies. Hydrogenated CDC maintained relatively high capacitance: its C_{sp} , which we measured using CV at 10 mV s^{-1} , slightly decreased from 102 to 93 F g^{-1} . The Annealed CDC material also showed good capacitance retention and lost only 7 F g^{-1} (from 77 to 70 F g^{-1}) as a result of the cycling. This agrees with our prior finding, and shows improved electrochemical stability of defunctionalized CDCs [20]. However, the Aminated CDC material experienced a 13 F g^{-1} decreased as a result of cycling, which was in line with its coulombic efficiency drop.

Table 2

Summary of electrochemical performance of differently functionalized CDCs with $[\text{EMIm}^+][\text{TFSI}^-]$ electrolyte at room temperature, as determined using cyclic voltammetry and impedance spectroscopy.

Material	ESR (Ω)	C_{sp} , $10 \text{ mV s}^{-1}/\mu\text{F cm}^{-2}$ (F g^{-1})	Randles–Sevcik coefficient	Time constant (s)	Ionic impedance (Ω)	Film resistivity ($\Omega \text{ cm}$)
Annealed CDC	2.82	5.36 (79.3)	0.793	11.74	9.56	2.10 ± 0.3
Aminated CDC	2.22	4.30 (74.2)	0.688	17.36	9.61	9.32 ± 0.7
Hydrogenated CDC	2.04	5.75 (87.1)	0.802	18.21	6.06	9.30 ± 0.5

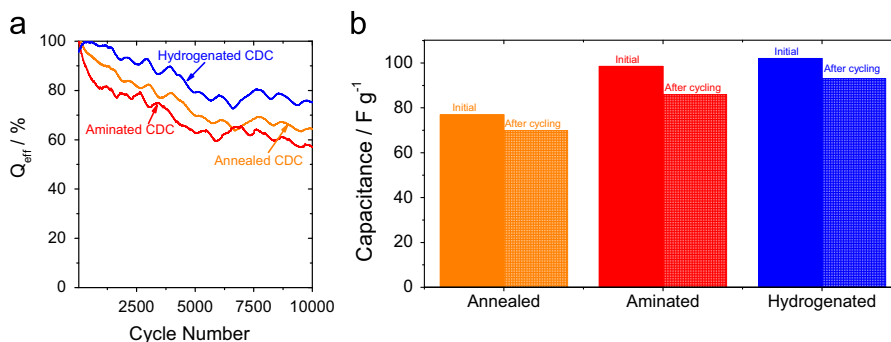


Fig. 5. (a) Coulombic efficiency comparison for annealed, aminated, and hydrogenated CDC after galvanostatic cycling at 5.0 A g^{-1} for 10,000 cycles. (b) Gravimetric capacitance of different CDCs before and after long-duration cycling, measured using cyclic voltammetry at 10 mV s^{-1} .

Long-duration CDC cycling did not show any signs of pseudocapacitive behavior. No new charge transfer processes in the cyclic voltammograms emerged after 10,000 cycles (shown in Fig. S6 in Supporting Information). Electrochemical breakdown of confined electrolyte and blockage of narrow pores with resulting decomposition products likely contributed most to supercapacitor degradation rather than oxidation or reduction of pore walls. Poorly reversible insertion of ions into narrow pores may also be partially responsible for this effect [47]. As compared to typically reported supercapacitors [48], the reported results significantly accelerated the cycling-induced device degradation with narrow pore diameters. However, the surface chemistries that improved ion mobilities and featured most favorable ion–surface interactions also correlated with greater capacitance retention over prolonged cycling. Subsequently, the ionic resistance during electrosorption has a long-term negative impact on the stability of $[\text{EMIm}^+][\text{TFSI}^-]$.

4. Conclusions

Hydrogenated and aminated pore surfaces have different effects on behavior of confined ions compared to defunctionalized, defect-free carbon pores. These findings implemented CDCs with nearly-identical porosities and completely decoupled the influence of ion confinement from intermolecular interactions between the electrode and $[\text{EMIm}^+][\text{TFSI}^-]$. Hydrogenated surfaces favorably attract ionic liquid and allow it to move more effectively through H-terminated pores than through defunctionalized channels. Conversely, we found that annealed, defect-free electrodes outperform nitrogen-functionalized CDCs. Ion electrosorption correlates with their mobility and dynamics as determined using quasi-elastic neutron scattering. Hydrogenated pores form favorable intermolecular interactions with confined electrolyte molecules, which, in turn, translate into improved capacitance, rate handling ability, and improved long-term cyclability. There is no contradiction between the slower microscopic dynamics observed in the quasi-elastic neutron scattering measurements and the better capacitance and rate handling abilities for the hydrogenated CDC. The former observation pertains to the time scale of tens to hundreds of picoseconds and underscores the stronger interaction of the hydrogenated CDC pore walls with ions. On the seconds time scale, all the ions are mobile, including those (the majority) immobilized on the pico-second time scale due to stronger interaction with the pore walls. For applications, the desirable characteristic is the ability of hydrogenated CDC to attract and retain the largest number of cations, as evidenced by quasi-elastic neutron scattering results.

These findings agree with our previous result, which stipulated that defects are beneficial to capacitance. However, these expanded observations demonstrate the selective influence of specific functional groups. While certain functionalities (hydrogen) are beneficial, others (amines) may negatively affect electrosorption and supercapacitor operation in this specific electrolyte. This finding is crucial for future electrode

materials designs and efforts to properly tailor high-end RTILs to appropriate porous carbons. CDC synthesis and structural analysis has, to date, primarily focused on the Cl_2 treatment conditions and viewed post-chlorination annealing with H_2 or NH_3 as an interchangeable step to remove leftover chlorine. This step can no longer be relegated to an afterthought, as the resulting chemical composition of CDC significantly influences resulting electrochemical performance. The specific intermolecular interactions between $[\text{EMIm}^+][\text{TFSI}^-]$ and surface groups are likely unique for each electrode–electrolyte interfaces, and future experiments must use a combination of in-depth characterization and predictive simulations to optimize the pore structure of materials for supercapacitors.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found in the online version at <http://dx.doi.org/10.1016/j.pnsc.2015.11.007>.

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